INCH-POUND

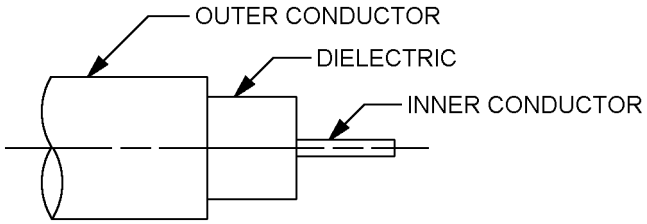
MIL-DTL-17/129E
26 July 2006
SUPERSEDING
MIL-C-17/129D
17 January 1992

DETAIL SPECIFICATION SHEET

# CABLES, RADIO FREQUENCY, COAXIAL, 0.250 (6.35 mm) DIAMETER, SEMIRIGID, 50 OHMS, M17/129-RG401 AND M17/129-00001

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.




| Inches | mm |
|---|---|
| .001 | 0.03 |
| .002 | 0.05 |
| .0641 | 1.682 |
| .209 | 5.31 |
| .250 | 6.35 |

* Metric conversions for table I

FIGURE 1. Configuration.

TABLE I. Description.

| Part or Identifying Number (PIN) | Inner conductor | Dielectric core | Outer conductor 1/ |
|---|---|---|---|
| M17/129-RG401 | Solid, silver-coated copper wire diameter, .0641 inch ± .0010 | Type F-1 Diameter, .209 inch ± .002 | Copper tubing diameter, .250 inch ± .001 |
| M17/129-00001 | Same as above | Same as above | Copper tubing diameter, .250 inch ± .002, -.001 Tin plated in accordance with ASTM B545, .0003 minimum inch thick 2/ |

1/ Welded outer conductor in accordance with ASTM B447 and MIL-DTL-17 is optional.
2/ The outer diameter dimensions is after plating.



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Engineering information:

Continuous working voltage: 3,000 V rms, maximum.

Operating frequency: 18 GHz, maximum.

Velocity of propagation: 69.5 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -40ºC to + 90ºC.

Inner conductor properties:

DC resistance (maximum at 20ºC): .257 ohms per 100 feet.

Elongation: 25 percent, minimum, per 10-inch specimen.

Engineering notes: This cable is generally useful in critical RF performance applications (see connector series TNC in accordance with MIL-PRF-39012). This cable is generally manufactured in 20-foot lengths. Different lengths are available.

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:

Eccentricity: 7.5 percent, maximum.

Adhesion of conductors:

Inner conductor to core: 4 pounds minimum; 100 pounds, maximum.

Outer conductor to core: 4 pounds minimum.

Aging stability: Not applicable.

Stress crack resistance: Not applicable.

Outer conductor integrity: Four specimens, approximately 2 feet long, shall be cut from the sample unit. The specimens shall be suspended in a heat chamber without touching one another or the walls of the chamber and conditioned for 1 hour 30 minutes minimum at the specified maximum operating temperature ±5º C. Heated air shall be circulated so as to maintain a uniform test temperature. After the conditioning period, the specimens shall be removed from the heat chamber and conditioned at room ambient temperature for 4 hours minimum. The specimen shall then have one end immersed into molten solder to a depth of 0.5 inch minimum for 15 seconds minimum. The molten solder temperature shall be +230°C minimum. After the conditioning period, the specimens shall be removed from the molten solder and conditioned at room ambient temperature for 1 hour minimum. Examine the specimens for cracks, flaws, or other damage in the outer conductor material.



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Cold bend: Not applicable.

Dimensional stability: +125ºC ± 5ºC.

Core from jacket: .015 inch, maximum.

Contamination: Not applicable.

Bendability: Mandrel diameter .75 inch, maximum.

Flammability: Not applicable.

Weight: 10.5 pounds per 100 feet, maximum (M17/129-RG401).
10.6 pounds per 100 feet, maximum (M17/129-00001).

Electrical:

Test frequency: 400 MHz to 18 GHz.

Spark test: Not applicable.

Voltage withstanding: 7,500 V rms, minimum.

Insulation resistance: Not applicable.

Corona extinction voltage: 3,000 V rms, minimum.

Characteristic impedance: 50 ohms ± 0.5.

Attenuation: See figure 2.

Structural return loss: See figure 3.

Capacitance: 29.6 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

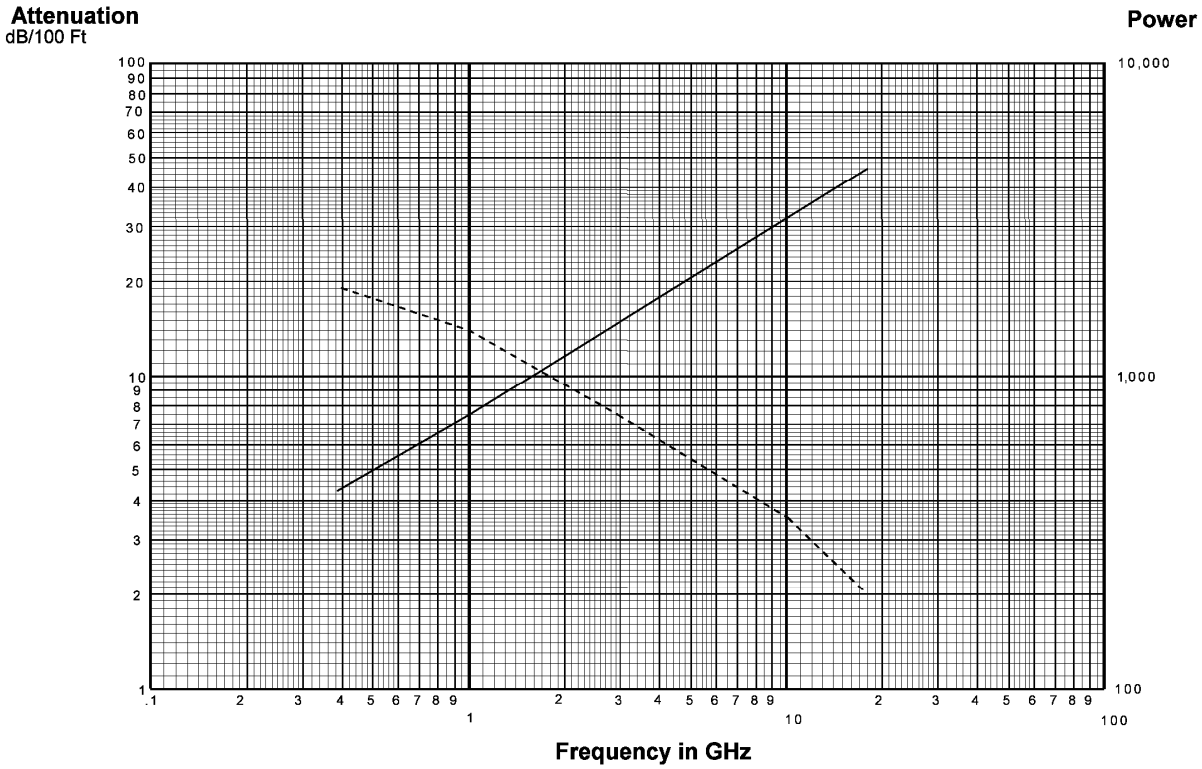
Time delay: Not applicable.

PIN: See table I.

Supersession data: See table II.



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Maximum attenuation: ———————— (Test requirements shall be a line indicted on graph)

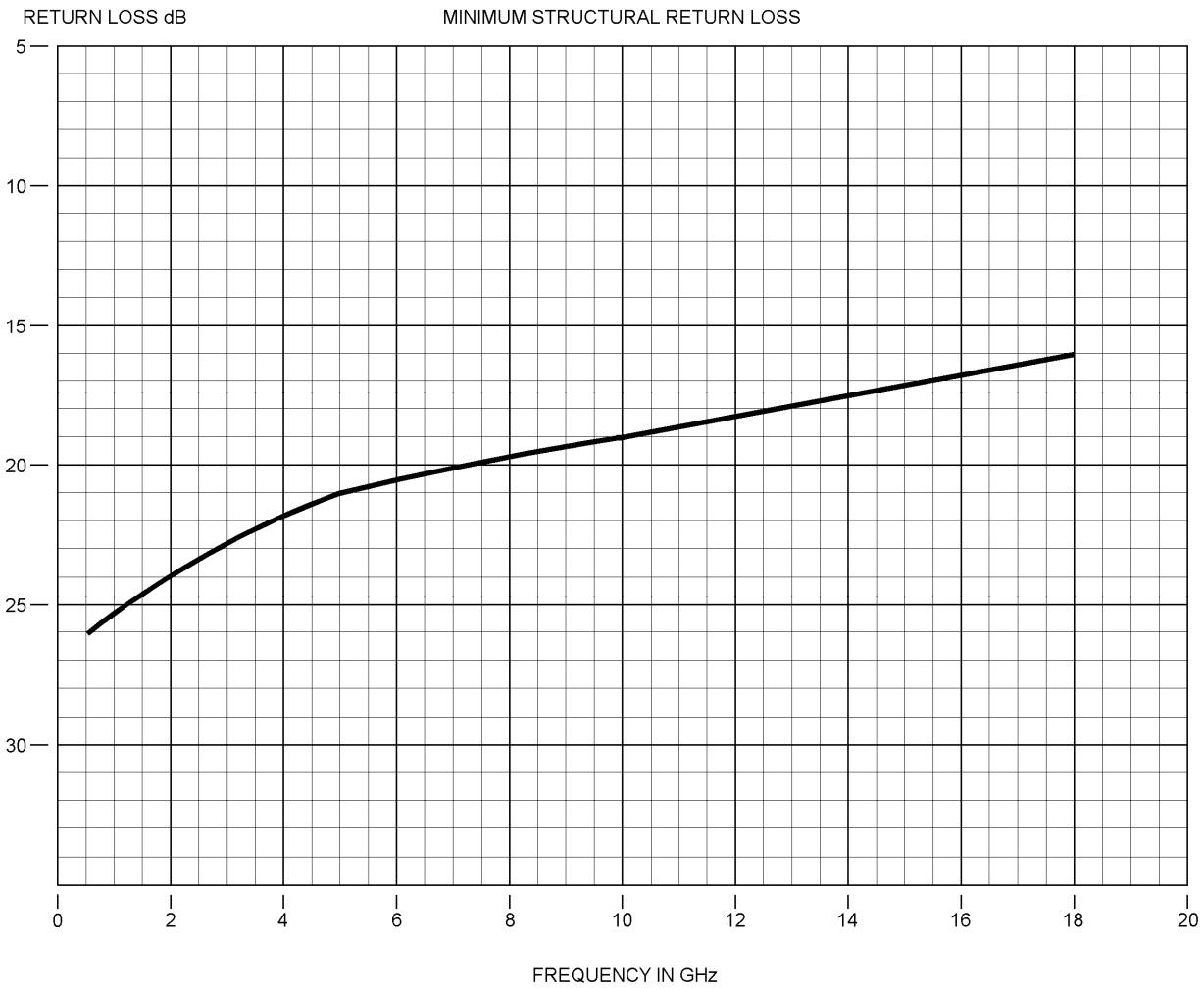
Maximum power - - - - - - - - - - - - - - - - -
(at 25°C at sea level)

| Attenuation | | Power |
|---|---|---|
| MHz | dB | Watts |
| 400 | 4.5 | 1900 |
| 1000 | 7.5 | 1400 |
| 3000 | 16 | 750 |
| 10000 | 33 | 350 |
| 18000 | 48 | 200 |

FIGURE 2. Power rating and attenuation.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:14Z
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| Structural Return Loss | |
|---|---|
| MHz | dB |
| 500 | 26 |
| 5000 | 21 |
| 10000 | 19 |
| 18000 | 16 |

FIGURE 3. Structural return loss.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:14Z
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| SWR | Reflection coefficient | Return loss dB |
|---|---|---|
| 17.3910 | .8913 | 1 |
| 8.7242 | .7943 | 2 |
| 5.8480 | .7079 | 3 |
| 4.4194 | .6310 | 4 |
| 3.5698 | .5623 | 5 |
| 3.0095 | .5012 | 6 |
| 2.6146 | .4467 | 7 |
| 2.3229 | .3981 | 8 |
| 2.0999 | .3548 | 9 |
| 1.9250 | .3162 | 10 |
| 1.7849 | .2818 | 11 |
| 1.6709 | .2512 | 12 |
| 1.5679 | .2239 | 13 |
| 1.4985 | .1995 | 14 |
| 1.4326 | .1778 | 15 |
| 1.3767 | .1585 | 16 |
| 1.3290 | .1413 | 17 |
| 1.2880 | .1259 | 18 |
| 1.2528 | .1122 | 19 |
| 1.2222 | .1000 | 20 |
| 1.1957 | .0891 | 21 |
| 1.1726 | .0794 | 22 |
| 1.1524 | .0708 | 23 |
| 1.1347 | .0631 | 24 |
| 1.1192 | .0562 | 25 |
| 1.1055 | .0501 | 26 |
| 1.0935 | .0447 | 27 |
| 1.0829 | .0398 | 28 |
| 1.0736 | .0355 | 29 |
| 1.0653 | .0316 | 30 |

FIGURE 3. Structural return loss - Continued.



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TABLE II. Cross reference of PIN.

| PIN | Superseded PIN or type designation |
|---|---|
| M17/129-RG401 | RG-401/U |
| M17/129-00001 | -- |

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-17, this document references the following:

ASTM B447
ASTM B545
MIL-PRF-39012

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2005-005)

Review activities:
Army - AT, AV, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99
DLA - IS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil .



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